ESP32-S3-Touch-LCD-3.5B

VERSION COMPATIBILITY NOTE

The new version of the ESP32-S3-Touch-LCD-3.5B includes the following hardware changes compared with the old version:

- Optimized camera power supply by adding a 4.7 Ω series resistor on the AVDD power line
- Optimized the PWR button detection circuit by adding a 100 kΩ pull-up resistor
- Adjusted the backlight driver circuit by switching to an AO3400 MOSFET and reserving a pull-down resistor
- Reserved ferrite beads and filter capacitors for the camera's `PCLK`, `SD_SCLK`, `I2S_SCLK`, and `LCD_QSPI_SCLK` signals
- Added a common mode inductor on the VBUS power line
- Reserved a conductive tape position on the PCB for shielding and grounding
- Optimized the RF section
- Upgraded the PCB from a 2-layer board to a 4-layer board

The old and new versions share the same programs, and the hardware changes above do not affect program usage.



This product is a high-performance, highly integrated microcontroller development board designed by Waveshare. It is equipped with a 3.5inch capacitive high-definition IPS screen, a highly integrated power management chip, a 6-axis IMU (3-axis accelerometer and 3-axis gyroscope), RTC, low-power audio codec chip and other peripherals, which are convenient for development and embedding into the product. No assembly or wiring is required, so it's easy to play with Xiaozhi AI.

If you are looking for:

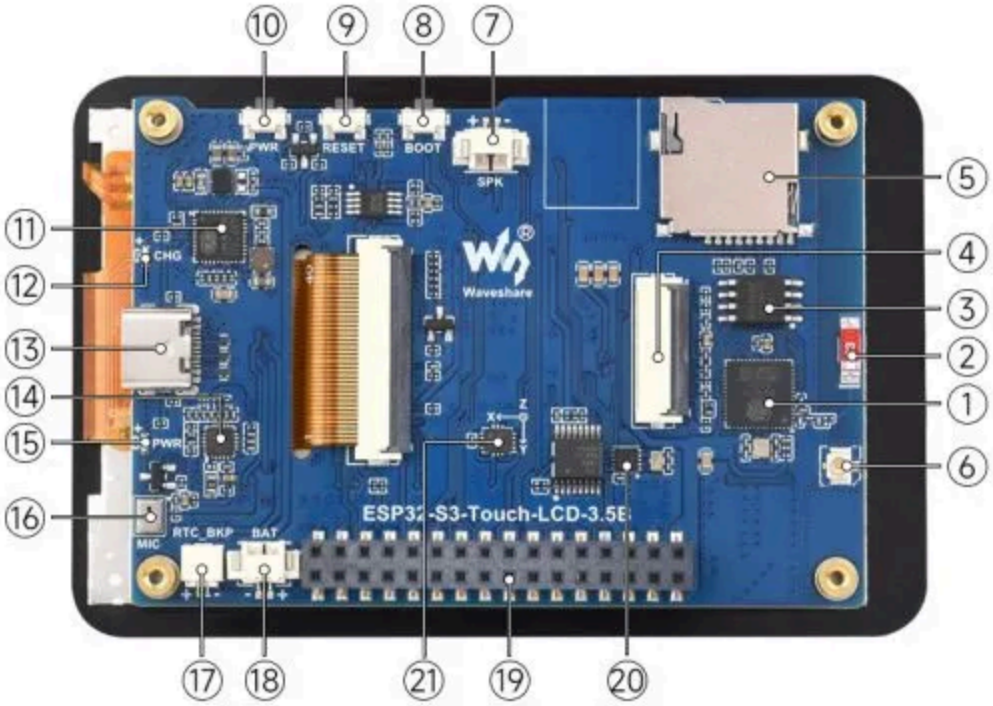
- **Coding with AI tools**: Click **Ask AI** and select an AI tool to ask questions about this page, or choose **Copy page** to copy its content and paste it into an AI tool for coding assistance.
- **Schematics, datasheets, and example code**: See Resources
- **Arduino development**: See Working with Arduino
- **ESP-IDF development**: See Working with ESP-IDF
- **Troubleshooting**: See Product FAQ or contact Technical Support

SKU	Product
31137	ESP32-S3-Touch-LCD-3.5B
31334	ESP32-S3-Touch-LCD-3.5B-C

Features

- Powered by the ESP32-S3R8 high-performance Xtensa 32-bit LX7 dual-core processor, with a main frequency of up to 240 MHz
- Supports 2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE) with an onboard antenna
- Built-in 512KB SRAM and 384 KB ROM, with stacked 8MB PSRAM and external 16MB Flash
- Features a Type-C interface, enhancing user convenience and device compatibility
- Built-in 3.5inch capacitive touch high-definition IPS display with a resolution of 320×480, 262K colors for clear color pictures
- Integrated AXS15231B display driver chip, communicating via QSPI and I2C interfaces
- Onboard QMI8658 6-axis IMU (3-axis accelerometer, 3-axis gyroscope), capable of detecting device orientation changes, step counting, and other functions
- Onboard PCF85063 RTC chip, powered by Lithium Batt through AXP2101 chip for uninterrupted power supply, with reserved RTC Batt pads for connecting a backup Batt, ensuring RTC function during the replacement of the main battery
- Onboard PWR and BOOT side buttons, configurable for custom function development
- Onboard 3.7V MX1.25 lithium battery charge/discharge interface
- Brings out I2C, UART, USB, and multiple GPIOs for external device connection and debugging, allowing flexible peripheral function configuration
- Onboard TF card slot, providing expanded storage, fast data transfer, and flexibility, suitable for data logging and media playback, simplifying circuit design
- Onboard camera interface, supporting mainstream cameras such as OV5640 and OV2640
- The benefits of using AXP2101 include efficient power management, support for multiple output voltages, charging and battery management functions, and optimization of battery life

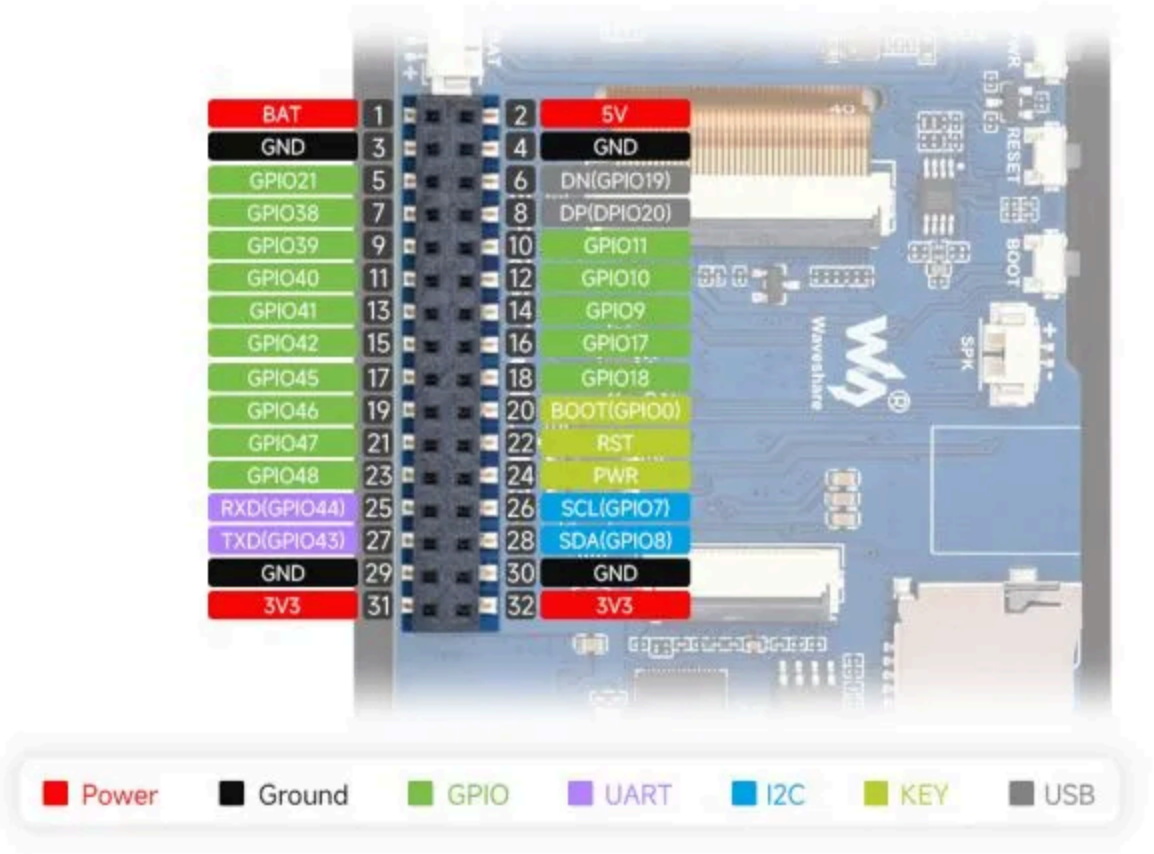
Onboard Resources



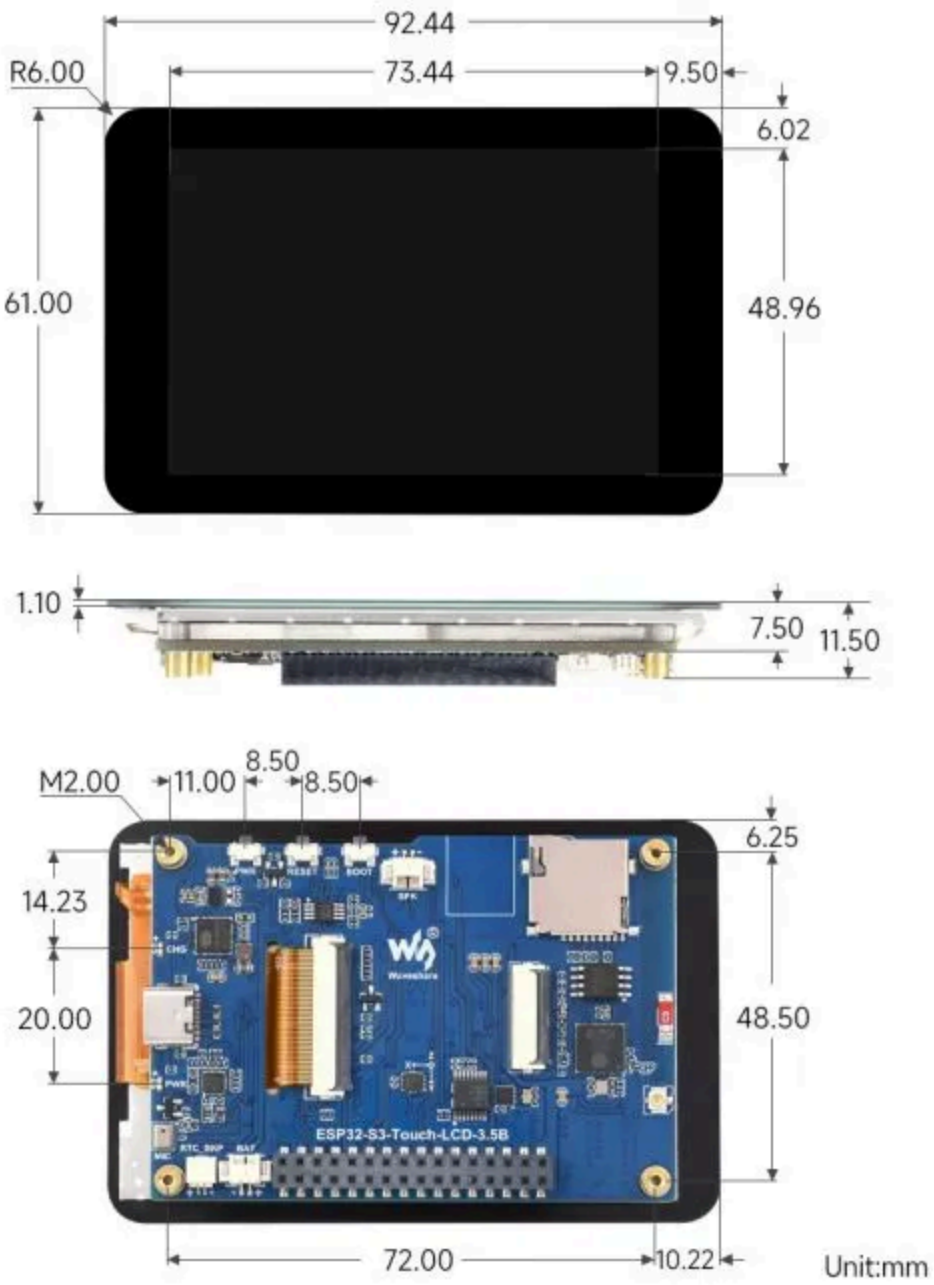
1. **ESP32-S3R8** Wi-Fi and Bluetooth SoC, 240MHz operating frequency, stacked 8MB PSRAM
2. **Onboard antenna**
3. **W25Q128JVSIQ** 16MB NOR-Flash
4. **Camera interface** Supports mainstream cameras such as OV5640 / OV2640
5. **TF card slot**
6. **IPEX1 connector** Switching to use the external antenna via resoldering an onboard resistor
7. **MX1.25 speaker header**
8. **BOOT button**
9. **RESET button**
10. **PWR button**
11. **AXP2101** Highly integrated power management chip
12. **Charging indicator**
13. **Type-C port**
14. **ES8311** Low-power audio codec chip
15. **Power indicator**
16. **Microphone**
17. **SH1.0 RTC battery header** For connecting a rechargeable RTC battery

18. **MX1.25 lithium battery header** MX1.25 2PIN connector for connecting a 3.7V lithium battery, supports charging and discharging
19. **2.54mm pitch GPIO header** Brings out available IO function pins for easy expansion
20. **PCF85063** RTC clock chip
21. **QMI8658** 6-axis IMU includes a 3-axis gyroscope and a 3-axis accelerometer

Pinout Definition



Dimensions


92.44
73.44
9.50
R6.00
6.02
61.00
48.96
1.10
7.50
11.50
M2.00
11.00
8.50
8.50
6.25
14.23
20.00
48.50
ESP32-S3-Touch-LCD-3.5B
72.00
10.22
Unit:mm



Development Methods

The ESP32-S3-Touch-LCD-3.5B supports two development frameworks: Arduino IDE and ESP-IDF, providing developers with flexible choices. You can select the appropriate development tool based on project requirements and personal preference.

Arduino generally offers a gentler learning curve and may be more approachable for beginners and hobbyists. ESP-IDF provides advanced tooling and finer control over system behavior, making it better suited to complex projects and applications with demanding performance requirements.

- **Arduino IDE** is a convenient, flexible, and easy-to-use open-source electronics prototyping platform. It requires minimal foundational knowledge, allowing for rapid development after a short learning period. Arduino has a huge global user community, providing a vast amount of open-source code, project examples, and tutorials, as well as a rich library ecosystem that encapsulates complex functions, enabling developers to implement various features rapidly. You can refer to the **Working with Arduino** to complete the initial setup, and the tutorial also provides related example programs for reference.

- **ESP-IDF**, short for Espressif IoT Development Framework, is a professional development framework launched by Espressif Systems for its ESP series of chips. It is based on C language development and includes compilers, debuggers, flashing tools, etc. It supports development via command line or integrated development environments (such as Visual Studio Code with the Espressif IDF plugin), which provides features like code navigation, project management, and debugging. We recommend using VS Code for development. For the specific configuration process, please refer to the **Working with ESP-IDF**. The tutorial also provides relevant example programs for reference.

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